

Cetus CLMM

Security Assessment

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01 — Executive Summary

Overview

Cetus Protocol engaged OtterSec to assess the `cetus-clmm-sui` program. This assessment was conducted between November 11th and November 15th, 2024. For more information on our auditing methodology, refer to Appendix B.

Key Findings

We produced 2 findings throughout this audit engagement.

We made a recommendation to modify the codebase for enhanced functionality and efficiency (OS-CLM-SUG-01) and suggested ensuring the pool is supplied with sufficient liquidity prior to creation (OS-CLM-SUG-00).

Scope

The source code was delivered to us in a Git repository at https://github.com/CetusProtocol/cetus-clmm-sui. This audit was performed against e3564d4.

A brief description of the program is as follows:

Name	Description
cetus-clmm-sui	The Cetus CLMM on the Sui blockchain optimizes decentralized exchange functionality by allowing liquidity providers to allocate their assets within specified price ranges, improving capital efficiency and yield opportunities.

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02 — Findings

Overall, we reported 2 findings.

We split the findings into **vulnerabilities** and **general findings**. Vulnerabilities have an immediate impact and should be remediated as soon as possible. General findings do not have an immediate impact but will aid in mitigating future vulnerabilities.



Severity	Count
CRITICAL	0
HIGH	0
MEDIUM	0
LOW	0
INFO	2

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03 — General Findings

Here, we present a discussion of general findings during our audit. While these findings do not present an immediate security impact, they represent anti-patterns and may result in security issues in the future.

ID	Description
OS-CLM-SUG-00	Users are allowed to create a pool by providing full-range liquidity but the function fails to validate the amounts of `coin_a` and `coin_b`. This oversight enables users to supply negligible amounts of one coin resulting in insufficient pool liquidity.
OS-CLM-SUG-01	Suggestions regarding inconsistencies in the codebase and ensuring adherence to coding best practices.

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Pool Creation with Insufficient Liquidity OS-CLM-SUG-00

Description

In the current implementation of `create_pool_v2_by_creation_cap` within `pool_create`, users have to provide full-range liquidity when creating a new pool. However, the function extracts `amount_a` and `amount_b` from the provided `coin_a` and `coin_b` without validating whether the amounts of `coin_a` and `coin_b` provided by the user are sufficient to contribute meaningful liquidity to the newly created pool.

>_ cetus_clmm/sources/pool_creator.move RUST

```
public fun create_pool_v2_by_creation_cap<CoinTypeA, CoinTypeB>(
    [...]
): Position {
    let (tick_lower_idx, tick_upper_idx) = full_range_tick_range(tick_spacing);
    let (amount_a, amount_b) = (coin::value(&coin_a), coin::value(&coin_b));
    let (position, coin_a, coin_b) =
        ↪  factory::create_pool_v2_by_creation_cap_<CoinTypeA,CoinTypeB>(
       [...]
    );
    [...]
}
```

Thus, it is possible to pass a small amount of one coin that only provides negligible liquidity to the pool, enabling the pool creation process to be abused to flood the system with pools with insufficient liquidity.

Remediation

Ensure the pool is supplied with sufficient liquidity prior to creation.

Patch

Resolved in 8d16203.

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Code Maturity OS-CLM-SUG-01

Description

1. Validate the `tick_spacing` parameter in `factory::add_allowed_pair_config` to ensure that only valid and meaningful tick spacings are allowed to be added to the allowed pair configuration.
2. `pool:check_pool_manager_role` should indeed utilize `is_pool_manager` to avoid duplicating the role-checking logic.

>_ *cetus_clmm/sources/config.move* RUST

```
// Check member is pool manager
public fun is_pool_manager(config: &GlobalConfig, member: address): bool {
    acl::has_role(&config.acl, member, ACL_POOL_MANAGER)
}

/// Check member has pool manager role
public fun check_pool_manager_role(config: &GlobalConfig, member: address) {
    assert!(acl::has_role(&config.acl, member, ACL_POOL_MANAGER), ENoPoolManagerPemission)
}
```

3. In `factory`, store `type_name::get<Coin>` in a variable within the `add_allowed_pair_config` to improving code efficiency, avoiding multiple calls to `type_name::get<Coin>`

Remediation

Implement the above-mentioned suggestions.

Patch

1. Issue #1 resolved in 8d16203.
2. Issue #2 resolved in 8d16203.
3. Issue #3 resolved in 8d16203.

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A — Vulnerability Rating Scale

We rated our findings according to the following scale. Vulnerabilities have immediate security implications. Informational findings may be found in the General Findings.

CRITICAL

Vulnerabilities that immediately result in a loss of user funds with minimal preconditions.

Examples:

- Misconfigured authority or access control validation.
- Improperly designed economic incentives leading to loss of funds.

HIGH

Vulnerabilities that may result in a loss of user funds but are potentially difficult to exploit.

Examples:

- Loss of funds requiring specific victim interactions.
- Exploitation involving high capital requirement with respect to payout.

MEDIUM

Vulnerabilities that may result in denial of service scenarios or degraded usability.

Examples:

- Computational limit exhaustion through malicious input.
- Forced exceptions in the normal user flow.

LOW

Low probability vulnerabilities, which are still exploitable but require extenuating circumstances or undue risk.

Examples:

- Oracle manipulation with large capital requirements and multiple transactions.

INFO

Best practices to mitigate future security risks. These are classified as general findings.

Examples:

- Explicit assertion of critical internal invariants.
- Improved input validation.

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B — Procedure

As part of our standard auditing procedure, we split our analysis into two main sections: design and implementation.

When auditing the design of a program, we aim to ensure that the overall economic architecture is sound in the context of an on-chain program. In other words, there is no way to steal funds or deny service, ignoring any chain-specific quirks. This usually requires a deep understanding of the program's internal interactions, potential game theory implications, and general on-chain execution primitives.

One example of a design vulnerability would be an on-chain oracle that could be manipulated by flash loans or large deposits. Such a design would generally be unsound regardless of which chain the oracle is deployed on.

On the other hand, auditing the program's implementation requires a deep understanding of the chain's execution model. While this varies from chain to chain, some common implementation vulnerabilities include reentrancy, account ownership issues, arithmetic overflows, and rounding bugs.

As a general rule of thumb, implementation vulnerabilities tend to be more "checklist" style. In contrast, design vulnerabilities require a strong understanding of the underlying system and the various interactions: both with the user and cross-program.

As we approach any new target, we strive to comprehensively understand the program first. In our audits, we always approach targets with a team of auditors. This allows us to share thoughts and collaborate, picking up on details that others may have missed.

While sometimes the line between design and implementation can be blurry, we hope this gives some insight into our auditing procedure and thought process.

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